

GUARD: Constructing Realistic Two-Player Matrix and Security Games for Benchmarking Game-Theoretic Algorithms

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Game Representations and Propositions

Normal (Matrix) Form Games (NFGs)

Player 1 chooses from n actions, Player 2 from m actions, and the payoff is determined from utility matrices $A, B \in \mathbb{R}^{n \times m}$. Their strategy spaces are given by simplexes Δ^n and Δ^m where:

$$\Delta^n = \{x \in \mathbb{R}_+^n \mid \sum_{i=1}^n x_i = 1\}$$

In a zero-sum game, $A = -B$, while in a general-sum game, $A \neq -B$. The best responses of Player 2 is the set $BR_2(x) = \operatorname{argmax}_{y \in [m]} x^T B y$

Schedule Form Security Games (SFGs)

Defenders establish coverage over a set of targets by assigning “schedules” of simultaneously coverable targets to defender resources. **SFGs are convertible to NFGs** by action matrix expansion.

Strong Stackelberg Eq. (SSE) in General Sum Games

In 2-player general-sum games, a Stackelberg equilibrium (x^*, y^*) can be formulated as a bilevel problem:

$$x^* = \operatorname{argmax}_{x \in \Delta^n} x^T A y^*(x), \text{ where } y^*(x) \in \operatorname{argmax}_{y \in [m]} x^T B y$$

Limitations of Benchmarking on Randomized Games

Many game-theoretic algorithms are **commonly benchmarked on randomized utility matrices** with uniform random payoffs.

Proposition 1: Player 1 achieves a near-optimal NFG payoff using a pure (single action) or extremely sparse strategy.

Proposition 2: In SFGs, the defender needs only sparingly use a single resource to capture near-optimal utility.

Acknowledgments

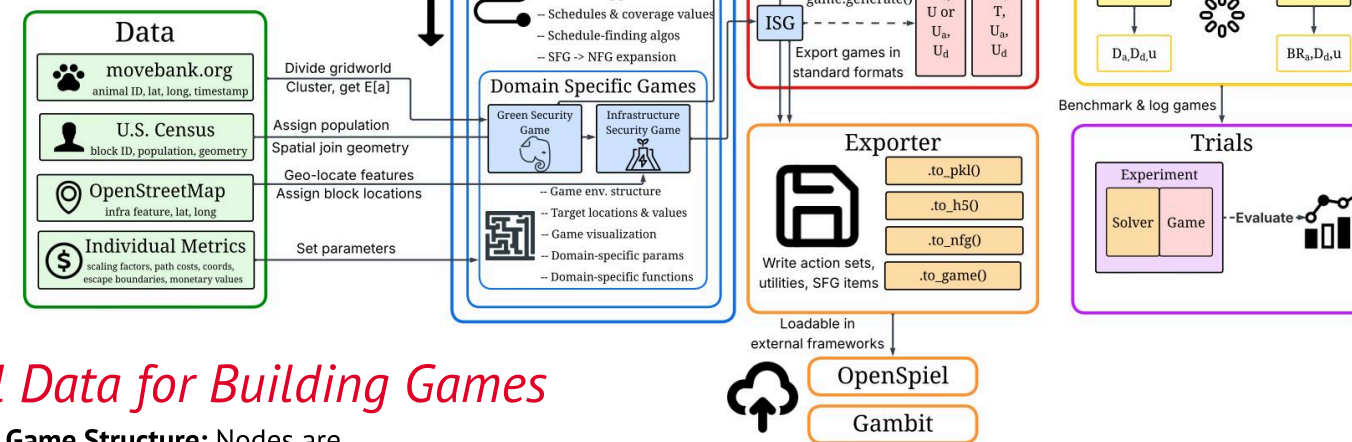
Elephant movement data provided by Movebank, urban graph and infrastructure data provided by OpenStreetMap and OSMNX, population data from CENSUS

GUARD Framework

Framework Structure

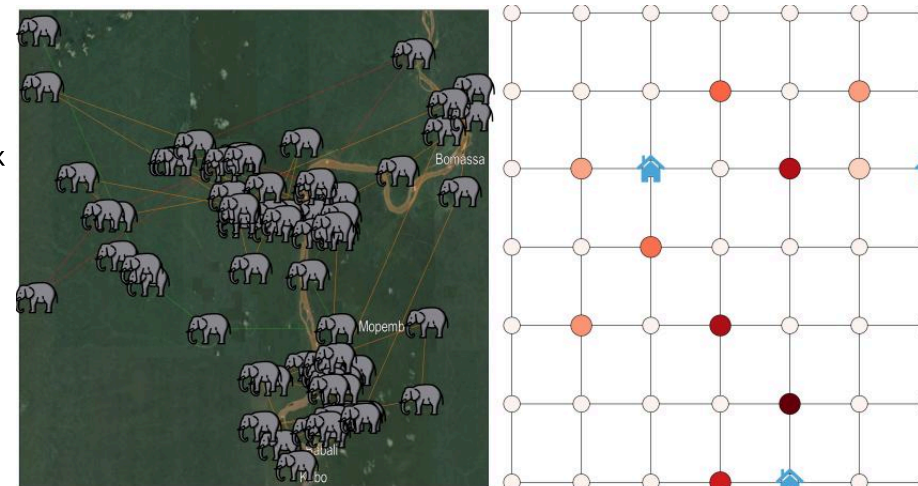
GUARD

Games Under Adversarial Realistic Domains

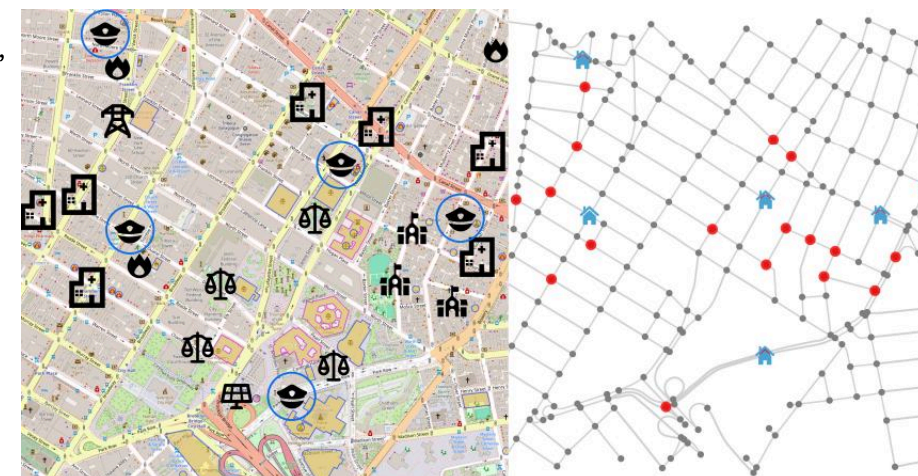


Real Data for Building Games

- **Graph Game Structure:** Nodes are intersections, edges are traversable paths.
 - GSGs: Bbox over nat. park coords. converted $\rightarrow$ gridworld $\rightarrow$ nx graph.
 - ISGs: osmnx bbox isolates a network of city streets $\rightarrow$ nx graph.
- **Target Locations:**
 - GSGs: Nodes closest to animal cluster centroids from movement datasets
 - ISGs: Nodes closest to OSM features (power, school, medical, etc.)
- **Target Values:**
 - GSGs: $E[\#A]$ scaled by avg. biodiversity \$ value for Def., avg. ivory \$ value for Att.
 - ISGs: Weight scaled by nearby block population ($w_{\text{school}} * 1000$ people, etc.)
- **Constraints:** Home bases, game length, etc.
 - GSGs: HB's are towns/ranger stations
 - ISGs: HB's are police stations
- **Options and Customizability**
 - Supports zero-sum & general-sum forms
 - Normal form & schedule-form games



Down-sampled elephant movements in Lobéké National Park and the corresponding GSG model.



Civil infrastructure in Manhattan's Chinatown and the corresponding ISG model.

Experiments & Results (GUARD vs. Rand.)

Iterative Algorithm Convergence

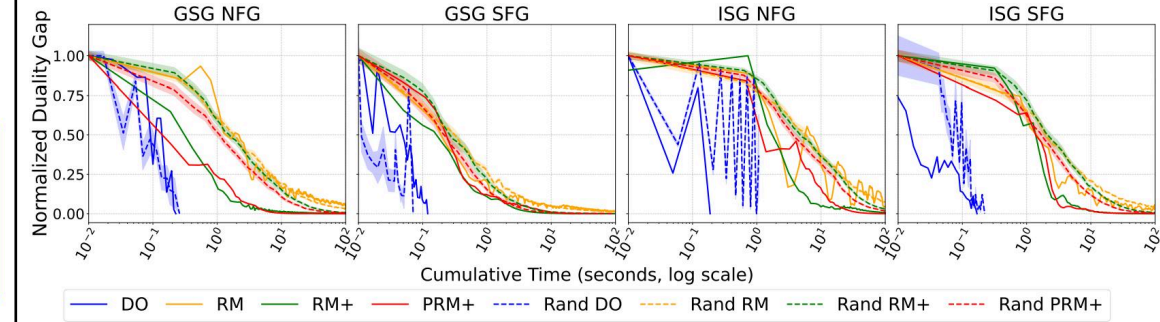


Figure 4: Convergence of iterative algorithms for different game types, formulations, and real vs. randomized (dashed lines with standard error bars) runs.

We found that **randomized zero-sum games exhibit more erratic D.O. convergence profiles and slower regret minimization** across RM variants compared to analogous realistic GUARD games, particularly in ISG environments.

Highlight: General Sum SSE. Profiles

Table 1: SSE Results for GSG and ISG settings. RT = randomized target values, RM = randomized matrix, RTS = randomized target values and schedules. u_d is max defender utility, **R** is runtime in seconds, and **S** is support of defender coverage strategy.

Form	Trial	GSG			ISG		
		u_d	R	S	u_d	R	S
Simple	Real	-0.390	0.015s	6	-0.429	0.030s	11
	RT	-0.298 $\pm$ 0.03	0.01s $\pm$ 0	5.0 $\pm$ 0.52	-0.214 $\pm$ 0.05	0.04s $\pm$ 0	7.5 $\pm$ 1.20
General	Real	-0.207	0.46s	9	-0.408	838s	14
	RM	-0.035 $\pm$ 0	0.52s $\pm$ 0.02	2 $\pm$ 0.15	-0.026 $\pm$ 0	541.5s $\pm$ 1.98	1.7 $\pm$ 0.15
	RT	-0.253 $\pm$ 0	0.35s $\pm$ 0.01	1.7 $\pm$ 0.26	-0.013 $\pm$ 0	116.2s $\pm$ 1.87	1 $\pm$ 0
	RTS	-0.270 $\pm$ 0	0.45s $\pm$ 0.03	1.4 $\pm$ 0.16	-0.032 $\pm$ 0	1.57 $\pm$ 0.04	1 $\pm$ 0

We found **significant SSE degeneracy in randomized general-sum SFGs**, with defender supports (strategy sizes) becoming extremely sparse and even pure, while trivially capturing optimal utility. GUARD general-sum games exhibited **richer strategies and more realistic utility capture**.

